Test of the overarrows package predefined commands

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1 \vv

```
\TestOverArrow*{\vv}
```

Test of \vv and \vv* macros

\vv for different math styles

\displaystyle	\textstyle	\scriptstyle	\scriptscriptstyle
$\vec{v}$	$\vec{v}$	$\vec{v}$	$\vec{v}$
$\overrightarrow{AB}$	$\overrightarrow{AB}$	$\overrightarrow{AB}$	$\overrightarrow{AB}$
$\overrightarrow{\mathrm{grad}}$	$\overrightarrow{\mathrm{grad}}$	$\overrightarrow{\mathrm{grad}}$	$\overrightarrow{\mathrm{grad}}$
$\overrightarrow{my\ long\ vector}$	$\overrightarrow{my\ long\ vector}$	$\overrightarrow{my\ long\ vector}$	$\overrightarrow{my\ long\ vector}$

\vv kerning

$$\vec{t}_{\vec{u}_{\vec{v}}} \qquad \vec{\imath}_0 \qquad \vec{v} = \vec{v}_x + \vec{v}_y + \vec{v}_z = v_x\vec{\imath} + v_y\vec{\jmath} + v_z\vec{k}$$

\vv* kerning

$$\vec{t}_{\vec{u}_{\vec{v}}} \qquad \vec{\imath}_0 \qquad \vec{v} = \vec{v}_x + \vec{v}_y + \vec{v}_z = v_x\vec{\imath} + v_y\vec{\jmath} + v_z\vec{k}$$

2 \overrightarrow

```
\TestOverArrow{\overrightarrow}
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\displaystyle	\textstyle	\scriptstyle	\scriptscriptstyle
$\overrightarrow{v}$	$\overrightarrow{v}$	$\overrightarrow{v}$	$\overrightarrow{v}$
$\overrightarrow{AB}$	$\overrightarrow{AB}$	$\overrightarrow{AB}$	$\overrightarrow{AB}$
$\overrightarrow{\text{grad}}$	$\overrightarrow{\text{grad}}$	$\overrightarrow{\text{grad}}$	$\overrightarrow{\text{grad}}$
$\overrightarrow{my\ long\ vector}$	$\overrightarrow{my\ long\ vector}$	$\overrightarrow{my\ long\ vector}$	$\overrightarrow{my\ long\ vector}$

3 \underrightarrow

```
\TestOverArrow{\underrightarrow}
```

\displaystyle	\textstyle	\scriptstyle	\scriptscriptstyle
$\underrightarrow{v}$	$\underrightarrow{v}$	$\underrightarrow{v}$	$\underrightarrow{v}$
$\underrightarrow{AB}$	$\underrightarrow{AB}$	$\underrightarrow{AB}$	$\underrightarrow{AB}$
$\underrightarrow{\text{grad}}$	$\underrightarrow{\text{grad}}$	$\underrightarrow{\text{grad}}$	$\underrightarrow{\text{grad}}$
$\underrightarrow{my\ long\ vector}$	$\underrightarrow{my\ long\ vector}$	$\underrightarrow{my\ long\ vector}$	$\underrightarrow{my\ long\ vector}$

4 \overleftarrow

```
\TestOverArrow{\overleftarrow}
```

\displaystyle	\textstyle	\scriptstyle	\scriptscriptstyle
$\overleftarrow{v}$	$\overleftarrow{v}$	$\overleftarrow{v}$	$\overleftarrow{v}$
$\overleftarrow{AB}$	$\overleftarrow{AB}$	$\overleftarrow{AB}$	$\overleftarrow{AB}$
$\overleftarrow{\text{grad}}$	$\overleftarrow{\text{grad}}$	$\overleftarrow{\text{grad}}$	$\overleftarrow{\text{grad}}$
$\overleftarrow{my\ long\ vector}$	$\overleftarrow{my\ long\ vector}$	$\overleftarrow{my\ long\ vector}$	$\overleftarrow{my\ long\ vector}$

5 \underleftarrow

```
\TestOverArrow{\underleftarrow}
```

\displaystyle	\textstyle	\scriptstyle	\scriptscriptstyle
$\underleftarrow{v}$	$\underleftarrow{v}$	$\underleftarrow{v}$	$\underleftarrow{v}$
$\underleftarrow{AB}$	$\underleftarrow{AB}$	$\underleftarrow{AB}$	$\underleftarrow{AB}$
$\underleftarrow{\text{grad}}$	$\underleftarrow{\text{grad}}$	$\underleftarrow{\text{grad}}$	$\underleftarrow{\text{grad}}$
$\underleftarrow{\textit{my long vector}}$	$\underleftarrow{\textit{my long vector}}$	$\underleftarrow{\textit{my long vector}}$	$\underleftarrow{\textit{my long vector}}$

6 \overleftrightarrow

```
\TestOverArrow{\overleftrightarrow}
```

\displaystyle	\textstyle	\scriptstyle	\scriptscriptstyle
$\overleftrightarrow{v}$	$\overleftrightarrow{v}$	$\overleftrightarrow{v}$	$\overleftrightarrow{v}$
$\overleftrightarrow{AB}$	$\overleftrightarrow{AB}$	$\overleftrightarrow{AB}$	$\overleftrightarrow{AB}$
$\overleftrightarrow{\text{grad}}$	$\overleftrightarrow{\text{grad}}$	$\overleftrightarrow{\text{grad}}$	$\overleftrightarrow{\text{grad}}$
$\overleftrightarrow{\textit{my long vector}}$	$\overleftrightarrow{\textit{my long vector}}$	$\overleftrightarrow{\textit{my long vector}}$	$\overleftrightarrow{\textit{my long vector}}$

7 \underleftrightarrow

```
\TestOverArrow{\underleftrightarrow}
```

\displaystyle	\textstyle	\scriptstyle	\scriptscriptstyle
$\underleftrightarrow{v}$	$\underleftrightarrow{v}$	$\underleftrightarrow{v}$	$\underleftrightarrow{v}$
$\underleftrightarrow{AB}$	$\underleftrightarrow{AB}$	$\underleftrightarrow{AB}$	$\underleftrightarrow{AB}$
$\underleftrightarrow{\mathrm{grad}}$	$\underleftrightarrow{\mathrm{grad}}$	$\underleftrightarrow{\mathrm{grad}}$	$\underleftrightarrow{\mathrm{grad}}$
$\underleftrightarrow{my\ long\ vector}$	$\underleftrightarrow{my\ long\ vector}$	$\underleftrightarrow{my\ long\ vector}$	$\underleftrightarrow{my\ long\ vector}$

8 \overrightharpoonup

```
\TestOverArrow{\overrightharpoonup}
```

\displaystyle	\textstyle	\scriptstyle	\scriptscriptstyle
$\overrightharpoonup{v}$	$\overrightharpoonup{v}$	$\overrightharpoonup{v}$	$\overrightharpoonup{v}$
$\overrightharpoonup{AB}$	$\overrightharpoonup{AB}$	$\overrightharpoonup{AB}$	$\overrightharpoonup{AB}$
$\overrightharpoonup{\mathrm{grad}}$	$\overrightharpoonup{\mathrm{grad}}$	$\overrightharpoonup{\mathrm{grad}}$	$\overrightharpoonup{\mathrm{grad}}$
$\overrightharpoonup{my\ long\ vector}$	$\overrightharpoonup{my\ long\ vector}$	$\overrightharpoonup{my\ long\ vector}$	$\overrightharpoonup{my\ long\ vector}$

9 \underrightharpoonup

```
\TestOverArrow{\underrightharpoonup}
```

\displaystyle	\textstyle	\scriptstyle	\scriptscriptstyle
$\underrightharpoonup{v}$	$\underrightharpoonup{v}$	$\underrightharpoonup{v}$	$\underrightharpoonup{v}$
$\underrightharpoonup{AB}$	$\underrightharpoonup{AB}$	$\underrightharpoonup{AB}$	$\underrightharpoonup{AB}$
$\underrightharpoonup{\mathrm{grad}}$	$\underrightharpoonup{\mathrm{grad}}$	$\underrightharpoonup{\mathrm{grad}}$	$\underrightharpoonup{\mathrm{grad}}$
$\underrightharpoonup{my\ long\ vector}$	$\underrightharpoonup{my\ long\ vector}$	$\underrightharpoonup{my\ long\ vector}$	$\underrightharpoonup{my\ long\ vector}$

10 \overrightharpoondown

```
\TestOverArrow{\overrightharpoondown}
```

\displaystyle	\textstyle	\scriptstyle	\scriptscriptstyle
$\overrightharpoondown{v}$	$\overrightharpoondown{v}$	$\overrightharpoondown{v}$	$\overrightharpoondown{v}$
$\overrightharpoondown{AB}$	$\overrightharpoondown{AB}$	$\overrightharpoondown{AB}$	$\overrightharpoondown{AB}$
$\overrightharpoondown{\mathrm{grad}}$	$\overrightharpoondown{\mathrm{grad}}$	$\overrightharpoondown{\mathrm{grad}}$	$\overrightharpoondown{\mathrm{grad}}$
$\overrightharpoondown{my\ long\ vector}$	$\overrightharpoondown{my\ long\ vector}$	$\overrightharpoondown{my\ long\ vector}$	$\overrightharpoondown{my\ long\ vector}$

11 \underrightharpoondown

```
\TestOverArrow{\underrightharpoondown}
```

\displaystyle	\textstyle	\scriptstyle	\scriptscriptstyle
$\underrightharpoondown{v}$	$\underrightharpoondown{v}$	$\underrightharpoondown{v}$	$\underrightharpoondown{v}$
$\underrightharpoondown{AB}$	$\underrightharpoondown{AB}$	$\underrightharpoondown{AB}$	$\underrightharpoondown{AB}$
$\underrightharpoondown{\mathrm{grad}}$	$\underrightharpoondown{\mathrm{grad}}$	$\underrightharpoondown{\mathrm{grad}}$	$\underrightharpoondown{\mathrm{grad}}$
$\underrightharpoondown{my\ long\ vector}$	$\underrightharpoondown{my\ long\ vector}$	$\underrightharpoondown{my\ long\ vector}$	$\underrightharpoondown{my\ long\ vector}$

12 \overleftharpoonup

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\TestOverArrow{\overleftharpoonup}
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\displaystyle	\textstyle	\scriptstyle	\scriptscriptstyle
$\overleftharpoonup{v}$	$\overleftharpoonup{v}$	$\overleftharpoonup{v}$	$\overleftharpoonup{v}$
$\overleftharpoonup{AB}$	$\overleftharpoonup{AB}$	$\overleftharpoonup{AB}$	$\overleftharpoonup{AB}$
$\overleftharpoonup{\mathrm{grad}}$	$\overleftharpoonup{\mathrm{grad}}$	$\overleftharpoonup{\mathrm{grad}}$	$\overleftharpoonup{\mathrm{grad}}$
$\overleftharpoonup{my\ long\ vector}$	$\overleftharpoonup{my\ long\ vector}$	$\overleftharpoonup{my\ long\ vector}$	$\overleftharpoonup{my\ long\ vector}$

13 \underleftharpoonup

```
\TestOverArrow{\underleftharpoonup}
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\displaystyle	\textstyle	\scriptstyle	\scriptscriptstyle
$\underleftharpoonup{v}$	$\underleftharpoonup{v}$	$\underleftharpoonup{v}$	$\underleftharpoonup{v}$
$\underleftharpoonup{AB}$	$\underleftharpoonup{AB}$	$\underleftharpoonup{AB}$	$\underleftharpoonup{AB}$
$\underleftharpoonup{\mathrm{grad}}$	$\underleftharpoonup{\mathrm{grad}}$	$\underleftharpoonup{\mathrm{grad}}$	$\underleftharpoonup{\mathrm{grad}}$
$\underleftharpoonup{my\ long\ vector}$	$\underleftharpoonup{my\ long\ vector}$	$\underleftharpoonup{my\ long\ vector}$	$\underleftharpoonup{my\ long\ vector}$

14 \overleftharpoondown

```
\TestOverArrow{\overleftharpoondown}
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\displaystyle	\textstyle	\scriptstyle	\scriptscriptstyle
$\overleftharpoondown{v}$	$\overleftharpoondown{v}$	$\overleftharpoondown{v}$	$\overleftharpoondown{v}$
$\overleftharpoondown{AB}$	$\overleftharpoondown{AB}$	$\overleftharpoondown{AB}$	$\overleftharpoondown{AB}$
$\overleftharpoondown{\mathrm{grad}}$	$\overleftharpoondown{\mathrm{grad}}$	$\overleftharpoondown{\mathrm{grad}}$	$\overleftharpoondown{\mathrm{grad}}$
$\overleftharpoondown{my\ long\ vector}$	$\overleftharpoondown{my\ long\ vector}$	$\overleftharpoondown{my\ long\ vector}$	$\overleftharpoondown{my\ long\ vector}$

15 \underleftharpoondown

```
\TestOverArrow{\underleftharpoondown}
```

\displaystyle	\textstyle	\scriptstyle	\scriptscriptstyle
$\underleftharpoondown{v}$	$\underleftharpoondown{v}$	$\underleftharpoondown{v}$	$\underleftharpoondown{v}$
$\underleftharpoondown{AB}$	$\underleftharpoondown{AB}$	$\underleftharpoondown{AB}$	$\underleftharpoondown{AB}$
$\underleftharpoondown{\mathrm{grad}}$	$\underleftharpoondown{\mathrm{grad}}$	$\underleftharpoondown{\mathrm{grad}}$	$\underleftharpoondown{\mathrm{grad}}$
$\underleftharpoondown{my\ long\ vector}$	$\underleftharpoondown{my\ long\ vector}$	$\underleftharpoondown{my\ long\ vector}$	$\underleftharpoondown{my\ long\ vector}$

16 \overbar

```
\TestOverArrow*{\overbar}
```

Test of \overbar and \overbar* macros

\overbar for different math styles

\displaystyle	\textstyle	\scriptstyle	\scriptscriptstyle
$\overline{v}$	$\overline{v}$	$\overline{v}$	$\overline{v}$
$\overline{AB}$	$\overline{AB}$	$\overline{AB}$	$\overline{AB}$
$\overline{\mathrm{grad}}$	$\overline{\mathrm{grad}}$	$\overline{\mathrm{grad}}$	$\overline{\mathrm{grad}}$
$\overline{my\ long\ vector}$	$\overline{my\ long\ vector}$	$\overline{my\ long\ vector}$	$\overline{my\ long\ vector}$

\overbar kerning

$$\overline{t}_{\overline{u}_{\overline{v}}} \qquad \overline{\imath}_{0} \qquad \overline{v} = \overline{v}_x + \overline{v}_y + \overline{v}_z = v_x\overline{\imath} + v_y\overline{\jmath} + v_z\overline{k}$$

\overbar* kerning

$$\overline{t}_{\overline{u}_{\overline{v}}} \qquad \overline{\imath}_{0} \qquad \overline{v} = \overline{v}_x + \overline{v}_y + \overline{v}_z = v_x\overline{\imath} + v_y\overline{\jmath} + v_z\overline{k}$$

17 \underbar

```
\TestOverArrow*{\underbar}
```

Test of \underbar and \underbar* macros

\underbar for different math styles

\displaystyle	\textstyle	\scriptstyle	\scriptscriptstyle
$\underline{v}$	$\underline{v}$	$\underline{v}$	$\underline{v}$
$\underline{AB}$	$\underline{AB}$	$\underline{AB}$	$\underline{AB}$
$\underline{\mathrm{grad}}$	$\underline{\mathrm{grad}}$	$\underline{\mathrm{grad}}$	$\underline{\mathrm{grad}}$
$\underline{my\ long\ vector}$	$\underline{my\ long\ vector}$	$\underline{my\ long\ vector}$	$\underline{my\ long\ vector}$

\underbar kerning

$$\underline{t}_{\underline{u}_{\underline{v}}} \quad \underline{\imath}_0 \quad \underline{v} = \underline{v}_x + \underline{v}_y + \underline{v}_z = v_x\underline{\imath} + v_y\underline{\jmath} + v_z\underline{k}$$

\underbar* kerning

$$\underline{t}_{\underline{u}_{\underline{v}}} \quad \underline{\imath}_0 \quad \underline{v} = \underline{v}_x + \underline{v}_y + \underline{v}_z = v_x\underline{\imath} + v_y\underline{\jmath} + v_z\underline{k}$$